

Fourier Transform: Phase, Sampling, and Time–Frequency Localization

Signal Processing

The Fourier transform describes a signal in terms of its frequency components. This handout distinguishes three related ideas: the amplitude and phase of a component, the sampling of the time and frequency axes, and the localization of a signal in these domains.

1 Continuous Fourier transform

We use frequency f in hertz and angular frequency $\omega = 2\pi f$. Our transform convention is

$$\boxed{X(f) = \int_{-\infty}^{\infty} x(t)e^{-i2\pi ft} dt, \quad x(t) = \int_{-\infty}^{\infty} X(f)e^{i2\pi ft} df.} \quad (1)$$

Euler’s identity,

$$e^{i\theta} = \cos \theta + i \sin \theta,$$

connects the complex exponential representation with cosine and sine.

For a real signal,

$$X(-f) = X(f)^*,$$

where the asterisk denotes complex conjugation. Thus, positive and negative frequencies are related rather than independent.

2 Cosine–sine coefficients and amplitude–phase

At one nonzero frequency, a real sinusoidal component can be written as

$$x_{\omega}(t) = a \cos(\omega t) + b \sin(\omega t). \quad (2)$$

The coefficients a and b are two independent real numbers.

Using

$$\cos(\omega t - \phi) = \cos \phi \cos(\omega t) + \sin \phi \sin(\omega t),$$

the same component becomes

$$\boxed{x_{\omega}(t) = A \cos(\omega t - \phi)} \quad (3)$$

with

$$\boxed{a = A \cos \phi, \quad b = A \sin \phi, \quad A = \sqrt{a^2 + b^2}, \quad \phi = \text{atan2}(b, a).} \quad (4)$$

If $A = 0$, phase is undefined.

These are Cartesian and polar descriptions of the same coefficient vector: (a, b) gives its coordinates, while (A, ϕ) gives its length and direction.

Why are cosine and sine 90° apart?

The fixed basis functions satisfy

$$\sin(\omega t) = \cos\left(\omega t - \frac{\pi}{2}\right).$$

They are therefore *in quadrature*: their relative phase is 90°.

The phase ϕ describes the resulting sinusoid relative to the time origin. It does not change the relative phase of the fixed basis functions.

One could introduce separate shifts,

$$a \cos(\omega t + \alpha) + b \sin(\omega t + \beta),$$

but expanding gives

$$(a \cos \alpha + b \sin \beta) \cos(\omega t) + (-a \sin \alpha + b \cos \beta) \sin(\omega t).$$

Thus, separate shifts produce another cosine–sine combination; the four parameters are redundant.

Connection with a complex Fourier coefficient

With the synthesis exponential $e^{i\omega t}$,

$$x_\omega(t) = \operatorname{Re}\{C e^{i\omega t}\}, \quad C = a - ib.$$

Consequently,

$$|C| = A, \quad \arg C = -\phi.$$

The minus sign follows from our choice $x_\omega(t) = A \cos(\omega t - \phi)$.

For a Fourier series written using both positive and negative frequencies, the corresponding coefficients are

$$c_+ = \frac{a - ib}{2}, \quad c_- = \frac{a + ib}{2},$$

because

$$x_\omega(t) = c_+ e^{i\omega t} + c_- e^{-i\omega t}.$$

3 Sampling, frequency range, and frequency spacing

Use different symbols for sampling intervals and signal widths.

Symbol	Meaning	Units
δt	Time sampling interval	s
δf	DFT frequency-bin spacing	Hz
T	DFT record duration	s
f_N	Nyquist frequency	Hz
σ_t	Energy-based temporal spread	s
σ_f	Energy-based spectral spread	Hz

For N samples taken at $t_n = n\delta t$, $n = 0, \dots, N-1$, the DFT period is

$$T = N\delta t.$$

The elapsed time between the first and last samples is $(N-1)\delta t$; $N\delta t$ is the duration used in the DFT spacing relation.

The DFT frequency spacing is

$$\delta f = \frac{1}{N\delta t} = \frac{1}{T}, \quad N\delta t \delta f = 1. \quad (5)$$

The Nyquist frequency is

$$f_N = \frac{1}{2\delta t}. \quad (6)$$

To avoid aliasing, the sampled signal must have its spectral content restricted below this limit, normally through appropriate filtering.

These equations express different controls:

- Smaller δt increases the available frequency range.
- Longer observation duration T decreases the frequency-bin spacing.

The full DFT frequency range has width

$$N\delta f = \frac{1}{\delta t} = 2f_N.$$

When expressed using signed frequencies, this is a frequency interval of width $2f_N$, conventionally chosen around zero.

Example

With

$$\delta t = 0.001 \text{ s}, \quad N = 1000,$$

we obtain

$$T = 1 \text{ s}, \quad \delta f = 1 \text{ Hz}, \quad f_N = 500 \text{ Hz}.$$

Recording for 2 s at the same sampling interval gives $\delta f = 0.5 \text{ Hz}$, while f_N remains 500 Hz.

Frequency spacing versus resolving power

Frequency-bin spacing describes the DFT grid. The ability to distinguish nearby spectral components also depends on observation duration, window shape, noise, and the estimation method.

For a rectangular observation window, the spectral main-lobe width scales as $1/T$. Increasing the duration of actual observations can therefore improve frequency discrimination.

Appending zeros decreases the DFT bin spacing but adds no observations. It samples the existing spectrum more densely; it does not narrow the spectral main lobe or improve the resolving power of the data.

4 Fourier transform of a boxcar

Consider a unit-height boxcar of duration L :

$$x(t) = \begin{cases} 1, & |t| \leq L/2, \\ 0, & |t| > L/2. \end{cases}$$

We use L for the pulse duration to distinguish it from the DFT record duration T .

Its Fourier transform is

$$X(f) = \int_{-L/2}^{L/2} e^{-i2\pi ft} dt \tag{7}$$

$$= \left[\frac{e^{-i2\pi ft}}{-i2\pi f} \right]_{-L/2}^{L/2} \tag{8}$$

$$= \frac{e^{-i\pi fL} - e^{i\pi fL}}{-i2\pi f} \tag{9}$$

$$= \frac{\sin(\pi fL)}{\pi f}. \tag{10}$$

Define the normalized sinc function by

$$\text{sinc}(u) = \frac{\sin(\pi u)}{\pi u}, \quad \text{sinc}(0) = 1.$$

Then

$$\boxed{\text{rect}(t/L) \longleftrightarrow L \text{sinc}(fL).} \tag{11}$$

At $f = 0$, the limit gives $X(0) = L$, equal to the area of the boxcar.

The first spectral zeros occur at $f = \pm 1/L$, so the main-lobe width between these zeros is

$$\boxed{W_{f,\text{main}} = \frac{2}{L}.} \tag{12}$$

A longer boxcar produces a narrower main lobe. Its abrupt edges produce sidelobes whose amplitude decays as $1/|f|$.

Multiplication by a boxcar observation window convolves a signal's spectrum with this sinc function. This explains spectral broadening and leakage associated with finite rectangular records.

5 Fourier transform of a Gaussian

Let

$$x(t) = e^{-t^2/(2s^2)}, \quad s > 0.$$

Here s is the width parameter of the *amplitude* Gaussian; it is not yet the energy-based standard deviation σ_t .

Its transform is

$$X(f) = \int_{-\infty}^{\infty} e^{-t^2/(2s^2)} e^{-i2\pi ft} dt.$$

We can derive the result by differentiating with respect to f . Since

$$x'(t) = -\frac{t}{s^2} x(t),$$

we obtain

$$X'(f) = -i2\pi \int_{-\infty}^{\infty} t x(t) e^{-i2\pi ft} dt \quad (13)$$

$$= i2\pi s^2 \int_{-\infty}^{\infty} x'(t) e^{-i2\pi ft} dt. \quad (14)$$

Integration by parts gives

$$\int_{-\infty}^{\infty} x'(t) e^{-i2\pi ft} dt = i2\pi f X(f),$$

because the boundary term vanishes. Therefore,

$$X'(f) = -4\pi^2 s^2 f X(f).$$

Solving this differential equation,

$$X(f) = X(0) e^{-2\pi^2 s^2 f^2}.$$

The Gaussian integral gives

$$X(0) = \int_{-\infty}^{\infty} e^{-t^2/(2s^2)} dt = \sqrt{2\pi} s.$$

Thus,

$$\boxed{e^{-t^2/(2s^2)} \longleftrightarrow \sqrt{2\pi} s e^{-2\pi^2 s^2 f^2}}. \quad (15)$$

The transform of a Gaussian is another Gaussian. Writing the frequency amplitude as $C e^{-f^2/(2s_f^2)}$ gives

$$s f = \frac{1}{2\pi s}, \quad \boxed{s s_f = \frac{1}{2\pi}}.$$

These are amplitude-profile width parameters.

6 Time–frequency uncertainty

The uncertainty principle concerns the spread of the *signal's energy*, rather than the spacing of sampled axes.

For a nonzero finite-energy signal, define

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df,$$

where the equality follows from Parseval's theorem.

The energy centroids are

$$\bar{t} = \frac{1}{E} \int t |x(t)|^2 dt, \quad \bar{f} = \frac{1}{E} \int f |X(f)|^2 df.$$

The corresponding variances are

$$\sigma_t^2 = \frac{1}{E} \int (t - \bar{t})^2 |x(t)|^2 dt, \quad \sigma_f^2 = \frac{1}{E} \int (f - \bar{f})^2 |X(f)|^2 df. \quad (16)$$

All integrals above extend over the real line.

When these variances are finite, the Fourier uncertainty relation is

$$\boxed{\sigma_t \sigma_f \geq \frac{1}{4\pi}.} \quad (17)$$

In angular frequency, $\sigma_\omega = 2\pi\sigma_f$, so

$$\boxed{\sigma_t \sigma_\omega \geq \frac{1}{2}.}$$

A signal cannot have arbitrarily small energy spreads in both time and frequency.

The Gaussian reaches equality

For $x(t) = e^{-t^2/(2s^2)}$,

$$|x(t)|^2 = e^{-t^2/s^2},$$

so

$$\sigma_t = \frac{s}{\sqrt{2}}.$$

Similarly,

$$|X(f)|^2 = 2\pi s^2 e^{-4\pi^2 s^2 f^2},$$

giving

$$\sigma_f = \frac{1}{2\sqrt{2}\pi s}.$$

Therefore,

$$\boxed{\sigma_t \sigma_f = \frac{s}{\sqrt{2}} \frac{1}{2\sqrt{2}\pi s} = \frac{1}{4\pi}.} \quad (18)$$

This differs from the amplitude-width product $s s_f = 1/(2\pi)$ because squaring a Gaussian makes its profile narrower.

The boxcar has infinite spectral variance

For the boxcar,

$$\sigma_t = \frac{L}{\sqrt{12}}.$$

However,

$$f^2 |X(f)|^2 = \frac{\sin^2(\pi f L)}{\pi^2},$$

whose integral over all frequencies diverges. Therefore,

$$\sigma_f = \infty.$$

The sinc main lobe has finite width, but its energy-based frequency variance is infinite. Main-lobe width and standard deviation are different measures of spectral width.

7 Key distinctions

Relation	Interpretation
$f_N = 1/(2\delta t)$	Time sampling controls the available frequency range.
$\delta f = 1/T$	DFT record duration controls frequency-bin spacing.
$\sigma_t \sigma_f \geq 1/(4\pi)$	Signal energy cannot be arbitrarily localized in both domains.

A short pulse can be sampled very finely and placed in a long, zero-padded record. Its DFT bins can then be closely spaced, while its intrinsic spectrum remains broad.

Cosine and sine are fixed basis functions in quadrature. Amplitude and phase describe their resulting sinusoid. Sampling intervals describe the axes; energy spreads describe the signal.